Module-2

Derivatives and Gradient descent

Acknowledgement

Thanks to the following courses and corresponding researchers for making their teaching/research material online

- Mike X Cohen's Master Deep Learning course (most contents taken from this Udemy Course)
- AI Tools such as Chat GPT, Grok, etc
- Introduction to Deep Learning, University of Illinois at Urbana-Champaign
- Introduction to Deep Learning, Carnegie Mellon University
- Convolutional Neural Networks for Visual Recognition, Stanford University
- Natural Language Processing with Deep Learning, Stanford University
- And Many More .....

Derivatives

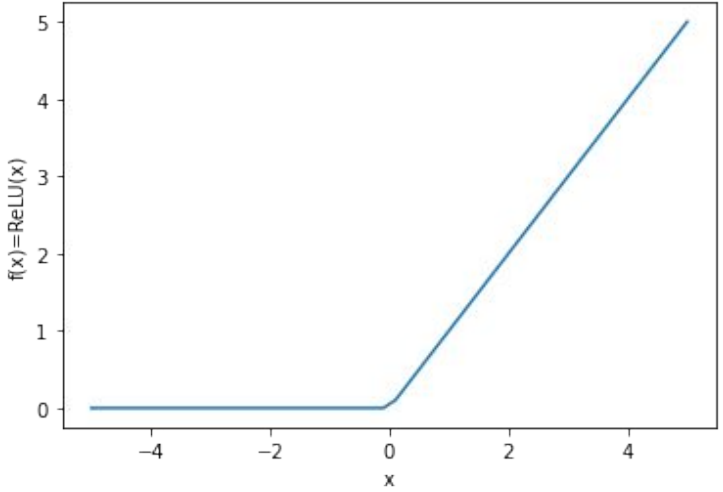
In this submodule, you will learn

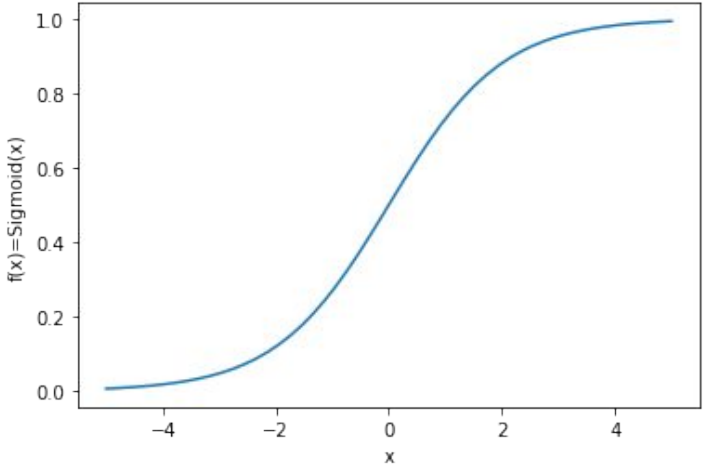
- Intuition of the derivative(“how things change”)
- How to compute the derivative of a polynomial.

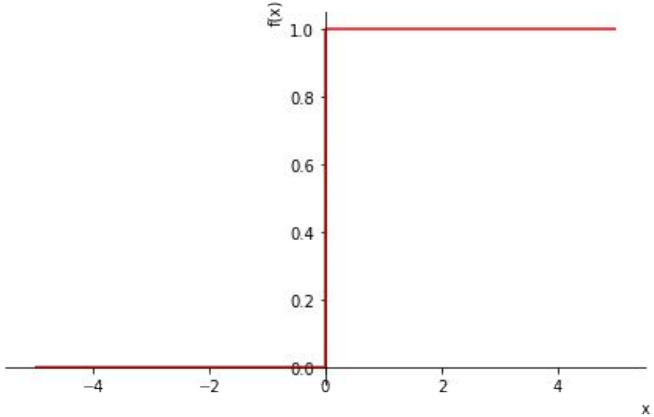
1

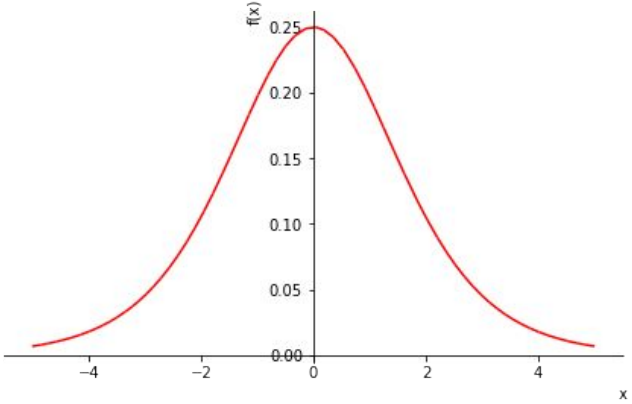
Derivatives

Derivative examples









Derivative of a polynomial function

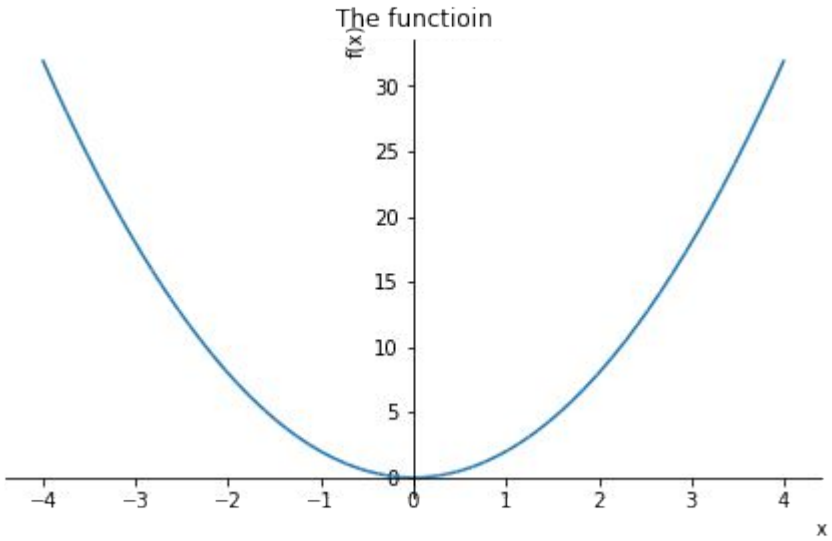
$$\boxed{\frac{d}{dx}x^2 = 2x^1}$$

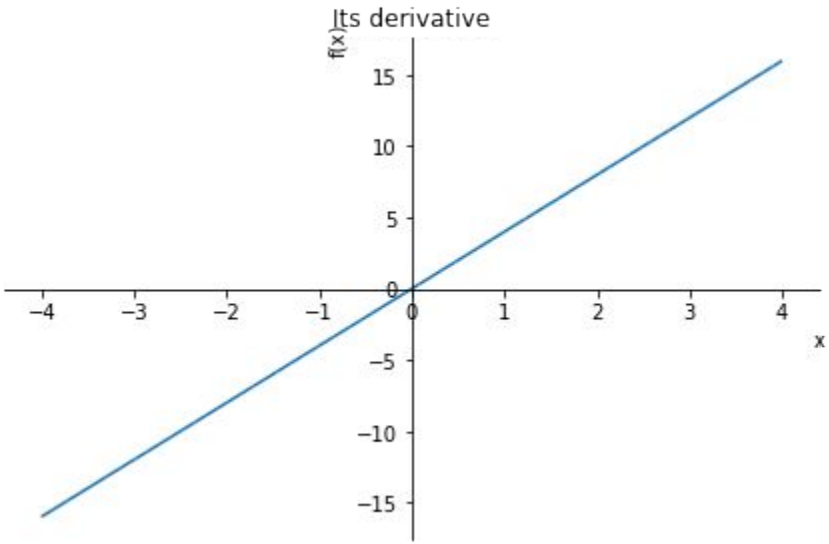
$$\boxed{\frac{d}{dx}3x^3 = 9x^2}$$

$$\boxed{\frac{d}{dx}ax^n = nax^{n-1}}$$

Derivative of a polynomial function

$$\frac{d}{dx}2x^2 = 4x^1$$





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Why bother with derivatives?

- Derivatives point in the direction of increases and decreases in a mathematical function.
- In DL, the goal(e.g. classification) is represented as an error function. Thus, the best solution is the point with the smallest error
- The derivative tells us which way to “move” in that error landscape in order to find the optimal solution.
- DL wouldn’t work without derivatives!

Derivatives

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- Intuition of the derivative(“how things change”)
- How to compute the derivative of a polynomial.

Quiz-1

1. What is the derivative of $f(x)=x^3$
   a. $3x^2$
   b. x^2
   c. $3x^3$
   d. x^3

2. The derivative of a constant function $f(x)=5$ is:
   a. 0
   b. 1
   c. 5
   d. Undefined

3. If $f(x)=\sin(x)$, then $f'(x)$ is
   a. $\cos(x)$
   b. $-\cos(x)$
   c. $\sin(x)$
   d. $-\sin(x)$

4. What is the derivative of the ReLU function $f(x)=\max(0,x)$?
   a. 1 for all x
   b. 0 for all x
   c. 0 if $x<0$, 1 if $x>0$
   d. Undefined everywhere

5. The derivative of the sigmoid function $\sigma(x)= 1/(1+e^{-x})$ is:
   a. $\sigma(x)^2$
   b. $\sigma(x)(1-\sigma(x))$
   c. e^x
   d. $1-\sigma(x))$

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2

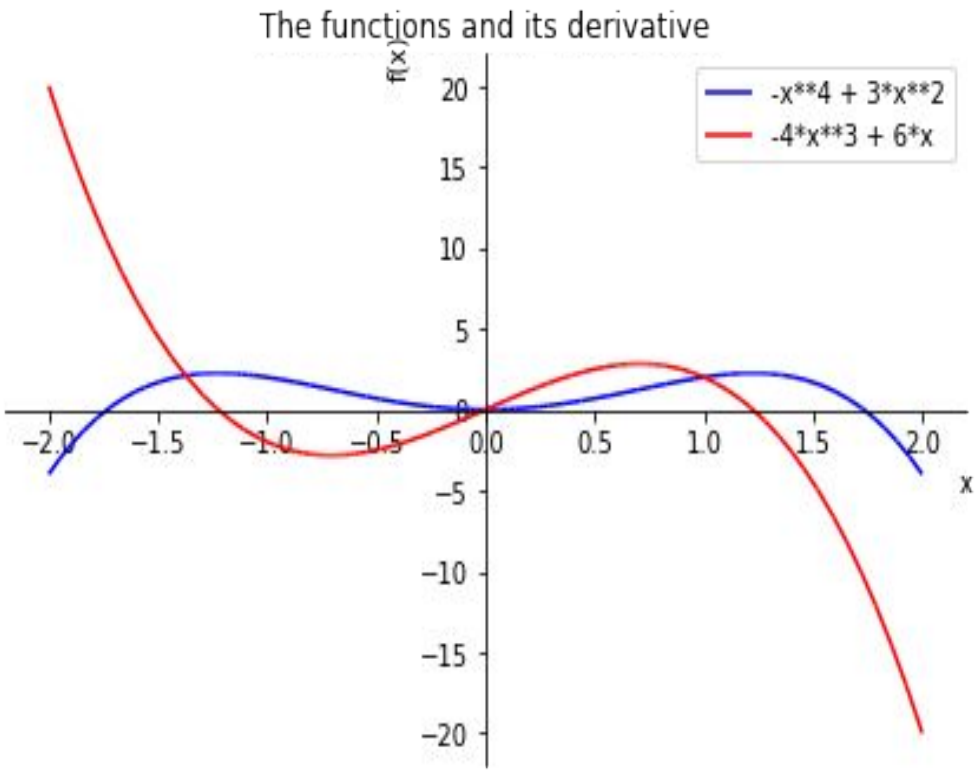
Derivatives find local minima

Derivatives find local minima

In this submodule, you will learn

- The main purpose of derivatives in gradient descent.
- What “minima” and “maxima” are
- How to distinguish local minima from local maxima

Minima and maxima



$$f(x) = -x^4 + 3x^2$$

$$\frac{df}{dx} = -4x^3 + 6x$$

$$-x^4 + 3x^2 = 0$$

$$x = 0, \pm\sqrt{3/2}$$

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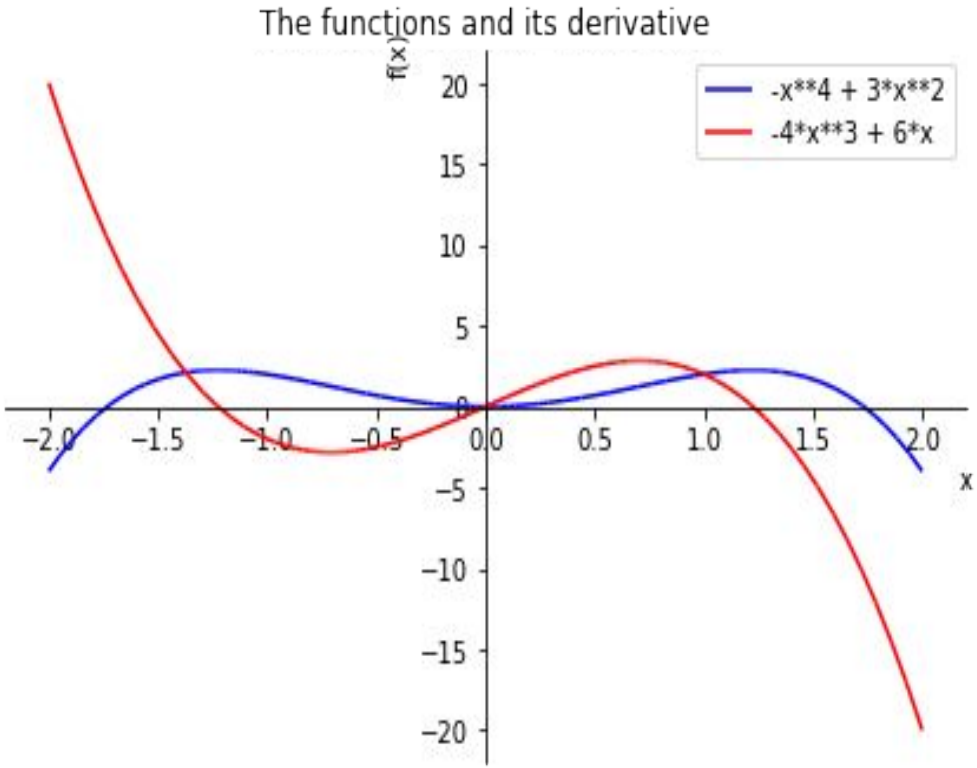
Minima and maxima

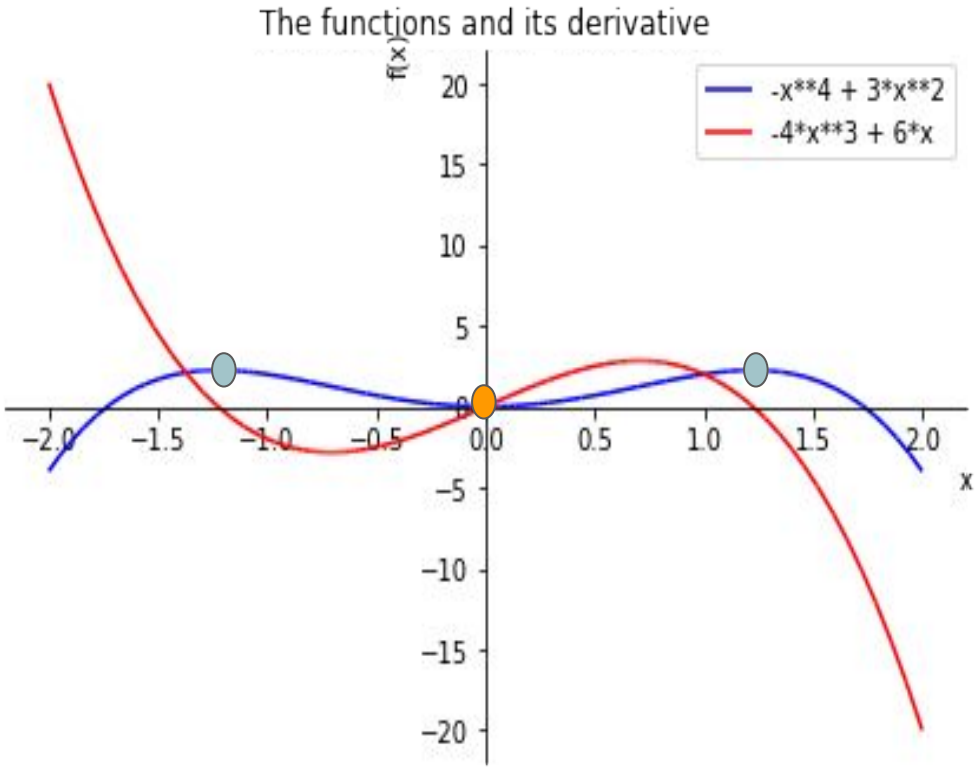


Minima and maxima have df=0

Minima have df<0 to the left, and df>0 to the right.

Maxima have df>0 to the left, and df<0 to the right

Minima and maxima



Minima and maxima have df=0

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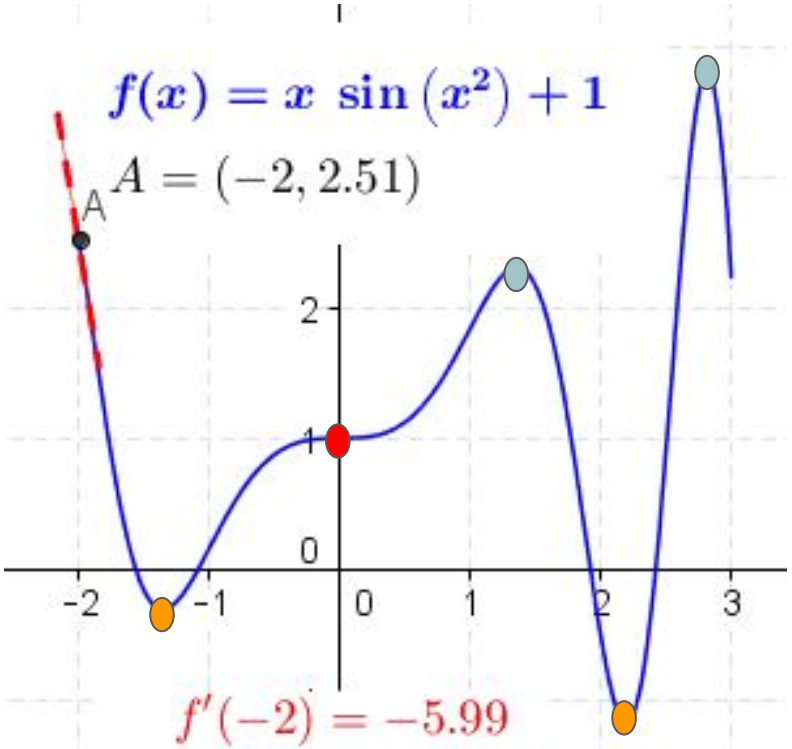
Maxima

Minima

Finding minima with derivatives



- Minima:
  df<0 to left, df>0 to right
- Maxima:
  df>0 to left, df<0 to right
- Constant:
  “Vanishing gradient”

Derivatives find local minima

In this submodule, you have learned about

- The main purpose of derivatives in gradient descent.
- What “minima” and “maxima” are
- How to distinguish local minima from local maxima

Quiz-2

1. Which of the following is true for finding a local minimum using the first derivative test?
   a. Derivative is constant on both sides of the point
   b. Derivative changes from positive to negative
   c. Derivative changes from negative to positive
   d. Derivative is zero everywhere

2. For the function $f(x) = x^3$, what can be said about $x = 0$?
   a. Local minimum
   b. Local maximum
   c. Neither minimum nor maximum
   d. Global minimum

3. Which method is used in optimization to find local minima using derivatives in higher dimensions?
   a. Gradient descent
   b. Binary search
   c. Linear regression
   d. Newton-Raphson method

4. If the first derivative of a function is always positive, then the function is:
   a. Constant
   b. Increasing
   c. Decreasing
   d. Concave

5. Consider $f(x) = x^2 + 4x + 5$. What is the local minimum point?
   a. x=-2
   b. x=0
   c. x=2
   d. No minimum exits

Quiz-2

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3

Derivatives: product and chain rules

Derivatives: product and chain rules

In this submodule, you will learn

- The “produce rule” of derivatives of multiple functions
- The Chain rule of derivatives of embedded functions

What to remember from this topic?

- Derivatives of “interacting” functions(multiplication and embedding) is unintuitive.
- In practice, libraries like pytorch, tensorflow etc., have routines that estimate complicated derivatives very efficiently and accurately.

The product rule of derivatives

$$(f+g)' = f' + g'$$

$$(f.g)' \neq f' \cdot g'$$

$$(f \cdot g)' = f' \cdot g + f \cdot g'$$

The chain rule of derivatives

$$\frac{df}{dx} f(g(x)) = f'(g(x))g'(x)$$

$$\frac{df}{dx} \left(x^2 + 4x^3\right)^5 = 5\left(x^2 + 4x^3\right)^4 \left(2x + 12x^2\right)$$

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Derivatives: product and chain rules

In this submodule, you have learned about

- The “produce rule” of derivatives of multiple functions
- The Chain rule of derivatives of embedded functions

Quiz-3

1. What is the derivative of $f(x) = (3x^2 + 2x)(x^3 - 5)$?
   a. $(6x+2)(3x^2-5)$
   b. $(6x+2)(x^3-5)+(3x^2+2x)(3x^2)$
   c. $(6x+2)(3x^2-15)$
   d. $(3x^2+2x)(3x^2-5)$

2. The chain rule is used when:
   a. Two functions are multiplied
   b. A function is raised to a constant
   c. One function is inside another
   d. A function is divided by another

3. What is the derivative of $f(x) = (2x + 5)^4$
   a. $8(2x+5)^3$
   b. $4(2x+5)^3$
   c. $4(2x+5)^2$
   d. None of the above

Quiz-3

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4

Overview of gradient descent

Overview of gradient descent

In this submodule, you will learn

- What “gradient descent” means and why we need it
- The gradient descent algorithm

Gradient descent:Theory and motivation

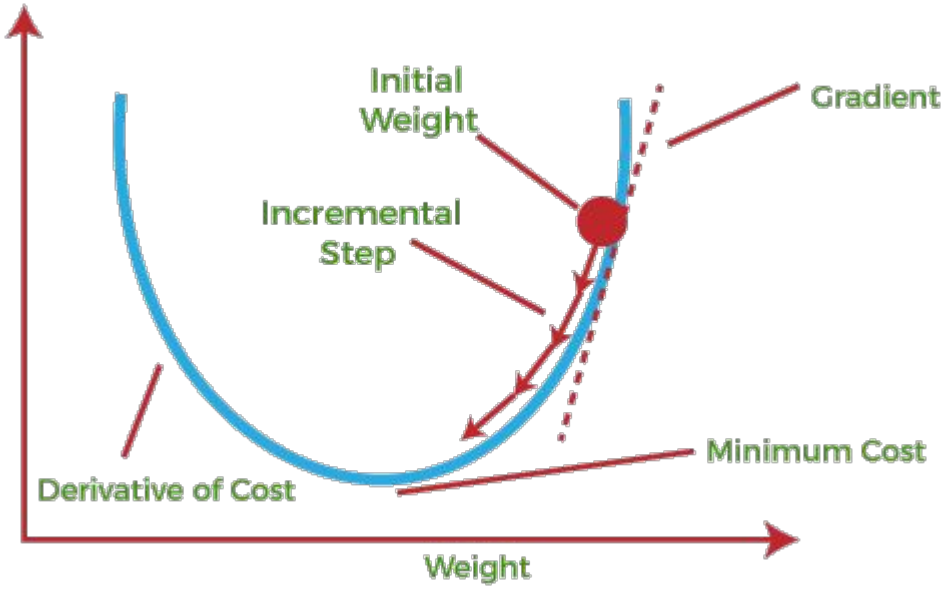
How deep learning models learn:

- Guess a solution.
- Compute the error(mistakes).
- Learn from mistakes and modify the parameters.

So: we need a mathematical description of the error “landscape” of the problem, and we need a way to find the minimum of the landscape.

Calculus to the rescue!

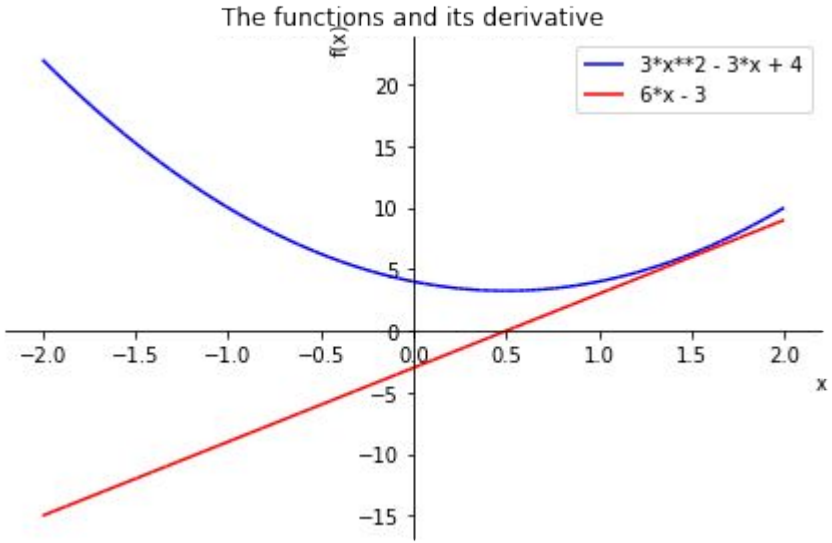
What is gradient descent?



The function and its derivative

$$f(x) = 3x^2 - 3x + 4$$

$$\frac{df}{dx} = 6x - 3$$



Gradient descent algorithms

- Initialize random guess of minimum
  - Loop over training iterations
    - Compute derivative at guess min
    - Updated guess min is itself minus derivative scaled by learning rate

Overview of gradient descent

In this submodule, you have learned about

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Quiz-4

1. What is the main goal of gradient descent?
   a. To find the global maximum of a function
   b. To find the roots of a function
   c. To minimize a cost or loss function
   d. To approximate the derivative of a function

2. In gradient descent, what does the "learning rate" control?
   a. Number of epochs
   b. Precision of the result
   c. Step size of the update
   d. Training data size

3. If the learning rate is too high, gradient descent might:
   a. Converge very slowly
   b. Skip over the minimum and diverge
   c. Not update weights at all
   d. Always converge to the global minimum

4. What is the gradient of a function at a local minimum point?
   a. Zero
   b. Positive
   c. Negative
   d. Undefined

5. In which direction does gradient descent move in parameter space?
   a. Opposite to the gradient
   b. Along the gradient
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5

What about local minima?

What about local minima?

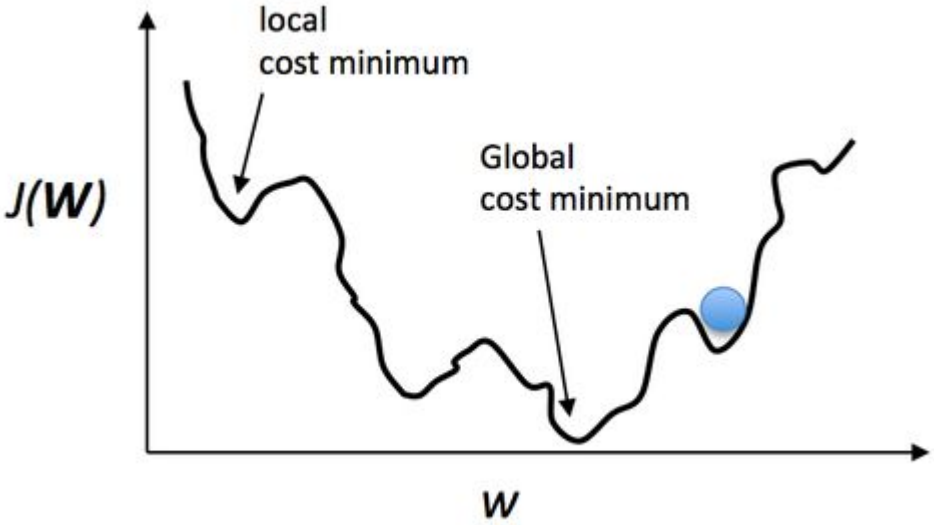
In this submodule, you will learn

- Potential problems with the g.d. algorithms.
- Why it’s hard to know whether this is serious issue.
- Two possible strategies to avoid local minima

Potential problems with gradient descent

- Gradient descent is guaranteed to go “downhill ”
- It is not guaranteed that to find the correct-or even the best solutions
- Gradient descent can go wrong if parameters are not set right for the particular error landscape.
- Error landscapes are impossible to visualize in >2D

Local vs. Global minima



Error landscape

But does it matter?

- The success of deep learning, in spite of the “problems” with gradient descent, remains a mystery.

- It is possible that there are many good solutions(many equally good local minima). This interpretation is consistent with the huge diversity of weight configurations that produce similar model performance.

- Another possibility is that there are extremely few local minima in high-dimensional space. The interpretation is consistent with the complexity and absurd dimensionality of DL Models.

What to do about it?

- When model performance is good, don’t worry about local minima.

- One possible solutions: Re-train the model many times using different random weights(different starting locations on the loss landscape) and pick the model that does best.

- Another possible solution: increase the dimensionality(complexity) of the model to have fewer local minima

What about local minima?

In this submodule, you have leaned about

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Quiz-5

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   a. The lowest value in the entire domain
   b. A point where the function increases in all directions
   c. A point where the function is lower than all nearby points
   d. A saddle point of the function

2. At a local minimum, the first derivative (gradient) of the function is:
   a. Always positive
   b. Always negative
   c. Zero
   d. Undefined

3. A local minimum is guaranteed to be a global minimum if the function is:
   a. Linear
   b. Convex
   c. Discrete
   d. Piecewise

4. What does the second derivative test determine at a critical point?
   a. Whether the point is a maximum, minimum, or saddle
   b. The exact value of the function
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Gradient descent in 1D

Gradient descent in 1D

In this submodule, you will learn

- See how gradient descent is implemented in code
- Learn how two important parameters (learning rate and training epochs) are implemented.
- Appreciate why gradient descent is not guaranteed to give the correct answer

The function its derivative

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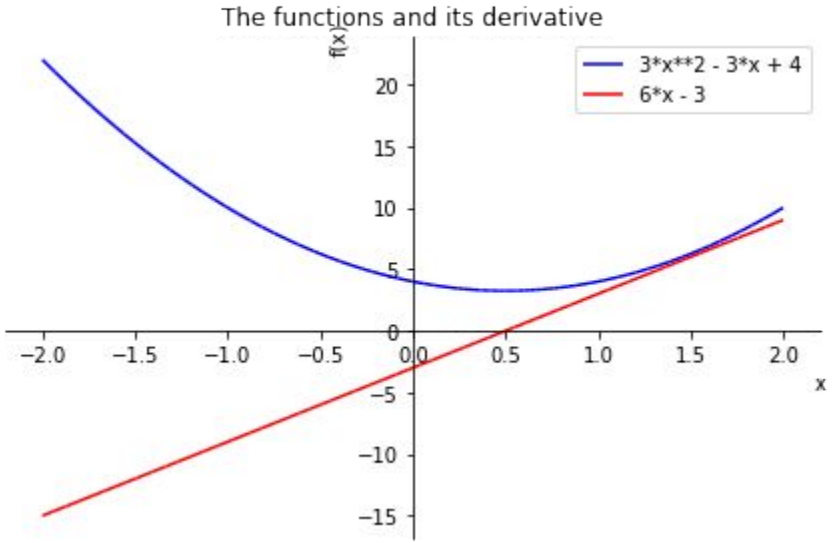

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- Convolutional Neural Networks for Visual Recognition, Stanford University
- Natural Language Processing with Deep Learning, Stanford University
- And Many More .....